

ENV 4120

Air Pollution and Hazardous Waste Control

Today's Topics

- Tropospheric ozone chemistry
- Expectation
 - Refresh basic chemical reaction kinetics on rate and rate constant
 - Understand the natural role of ozone, background ozone cycle in troposphere, photostationary state, role of ROG in tropospheric O₃ formation.

LA Smog



Beijing smog

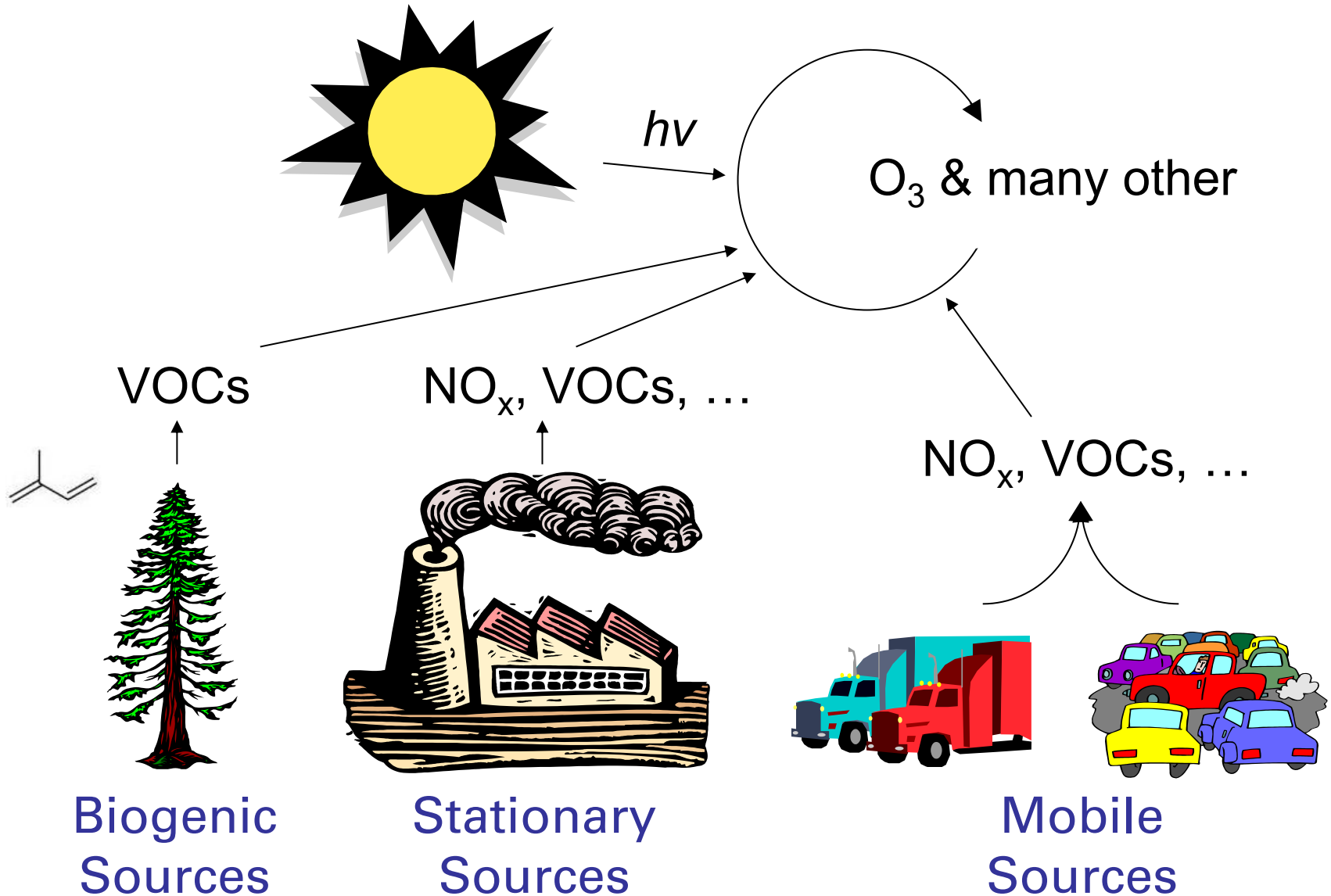


Smog in 20 minutes. Beijing, China. Jan 2nd, 2017
Credit: YouTube/Chas Pope

Smog

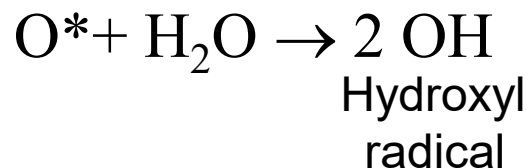
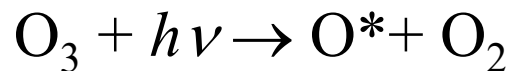
- Photochemical Smog (smoke+fog)
 - ‘Soup’ of gases and aerosols (primary + secondary)
 - Hazardous to health
 - Damage to materials, crops, visibility degradation, effects on climate and clouds
- There are many chemicals within smog
 - PM, O₃, HCs, NO_x, SO₂ etc
 - Ozone is a major component

photochemical smog



Background ozone chemistry

- O₃ do naturally occur in troposphere
- Approximately 10% of O₃ is in the troposphere
 - The rest is in stratospheric ozone layer
- Photolysis of O₃ leads to the production of hydroxyl radical (OH)
 - Tropospheric O₃ is important source
- Reaction with OH radical is the most important loss mechanism for almost all trace gases



Basic chemical kinetics: reaction rate

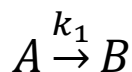
- How fast is the transformation from reactants to products: **Rates (*r*)**
 - Change in pollutant concentration over time

$$\text{unit: } \frac{\text{concentration}}{\text{time}}$$

- Rate laws:

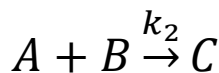
$$k_1: \text{rate constant. unit: } \frac{1}{s}$$

First order reaction
(unimolecular)



$$r = k_1[A] = \frac{d[B]}{dt} = -\frac{d[A]}{dt}$$

Second order reaction
(bimolecular)

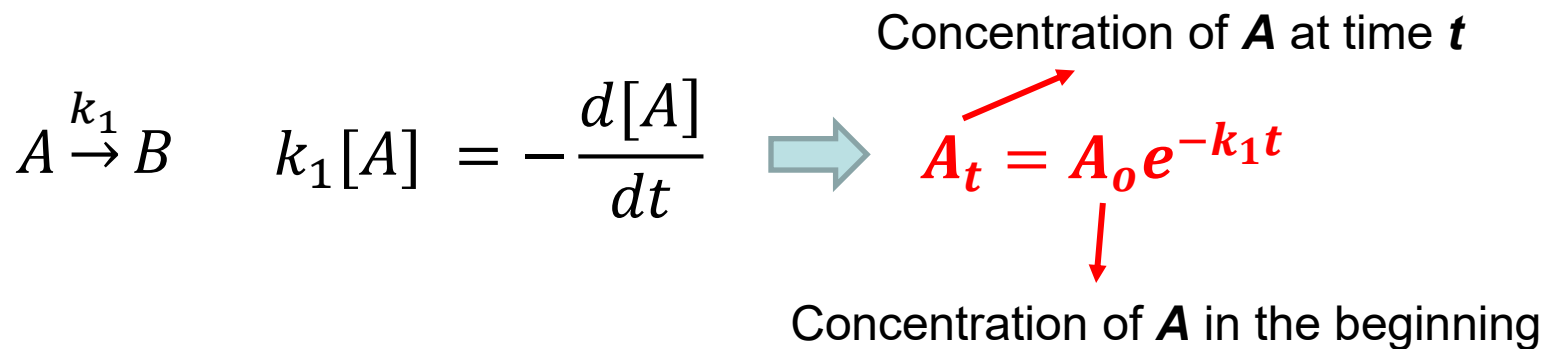


$$r = k_2[A][B] = \frac{d[C]}{dt} = -\frac{d[A]}{dt} = -\frac{d[B]}{dt}$$

$$k_2: \text{rate constant. unit: } \frac{\text{cm}^3}{\text{molecule} \cdot s} \text{ or } \frac{1}{\text{ppm} \cdot s}$$

Basic chemical kinetics: reaction rate

- For first order reaction

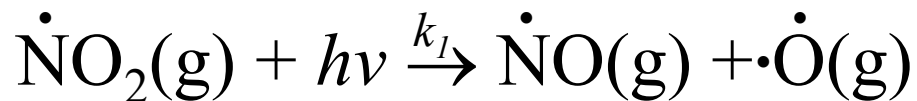


- Example: If $A_o = 95$ ppbv, $k_1 = 2 \times 10^{-3} \text{ sec}^{-1}$, what is A at 1 min?
- Half-life: time it takes for concentration of A to drop 50%

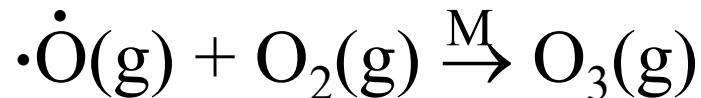
$$t_{1/2} = \frac{\ln(2)}{k}$$

Background ozone chemistry

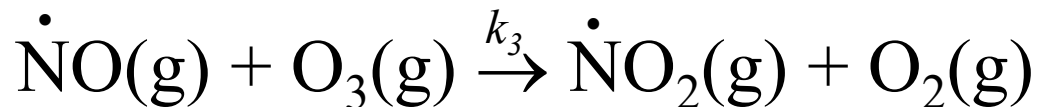
- Now consider a clean (background) atmosphere
- Three important reactions:



photolysis of NO_2
(production of O and NO)

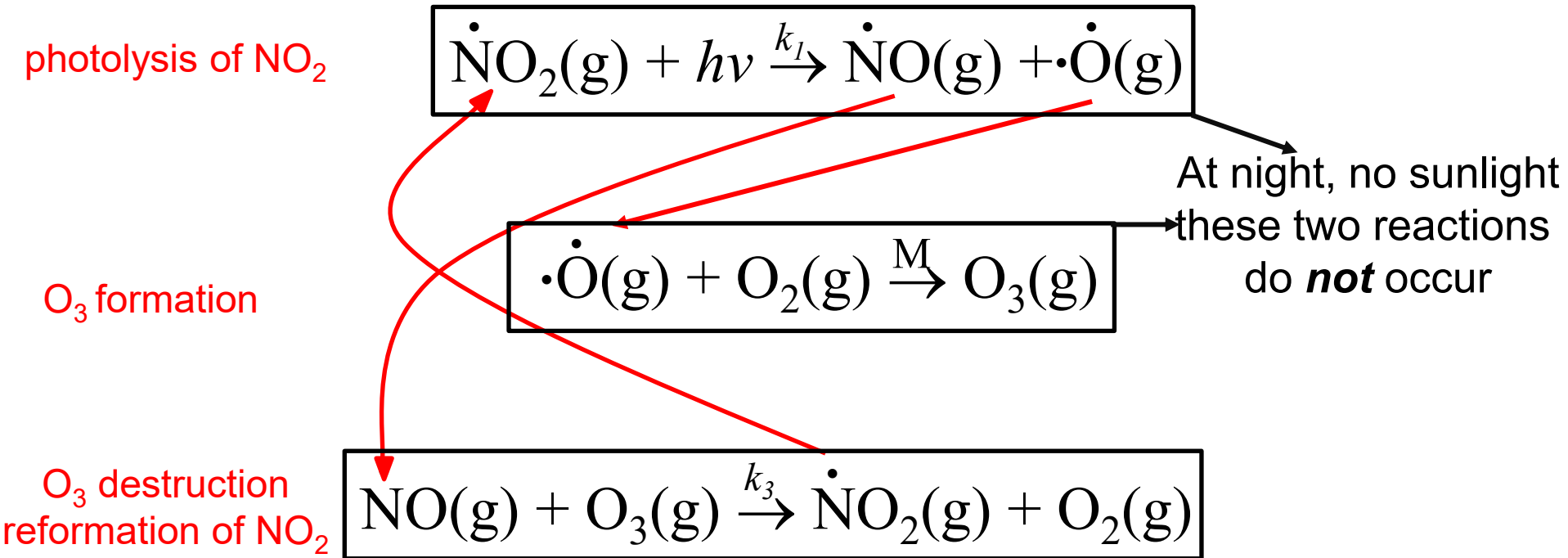


rxn of atomic oxygen with O_2 to
form O_3 (almost instantaneous)



rxn of NO with O_3 (ozone
destruction, reformation of NO_2)

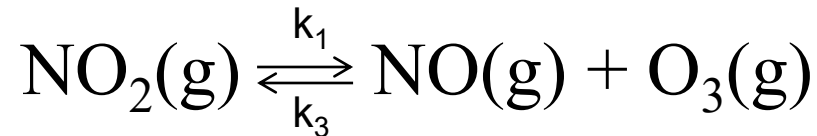
Background ozone chemistry



At night, this reaction still occur, lead to ozone destruction, especially near big NO emission sources: **NO_x titration**

Background ozone chemistry

- A simplify representation:



- Ozone going through a cycle of formation/removal
- At equilibrium: ***photostationary state (pss)***
 - Only in a few minutes with sunlight
- At this time, ozone concentration is given by:

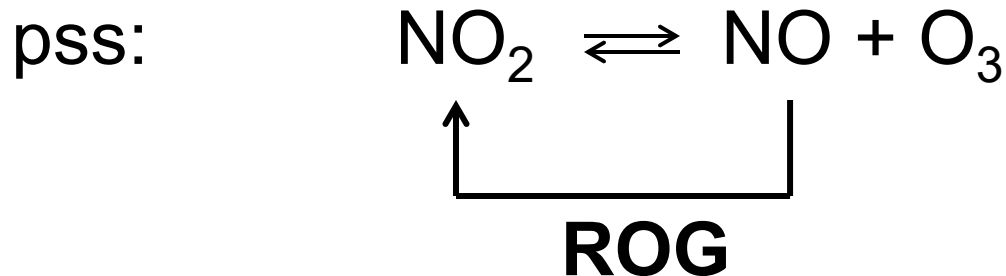
$$[\text{O}_3] = \frac{k_1 [\text{NO}_2]}{k_3 [\text{NO}]} \propto \frac{[\text{NO}_2]}{[\text{NO}]}$$

More NO₂, more ozone
More NO, less ozone

Background ozone chemistry

- Under clear sky, mid-latitude, mid afternoon conditions: $k_1/k_3 = 18$ ppbv
 - Typical value, real ratio do vary
- BUT, ozone concentration normally way higher than that estimated using k_1/k_3
- Also, ozone concentration not highest near NOx emission sources, bur rather downwind
 - Recall only a few minutes to reach pss

Reactive OrGanics (ROG)



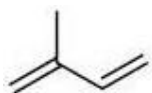
Role of reacting organics in smog formation:

- convert NO to NO₂, thereby increasing O₃ via pss
- produce aldehydes and peroxyacetyl nitrates (PANs)

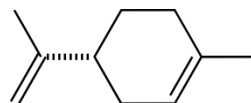
$$[O_3] = \frac{k_1 [NO_2]}{k_3 [NO]} \propto \frac{[NO_2]}{[NO]}$$

ROG sources

- Many volatile organic compounds (VOCs)
- Sources
 - anthropogenic
 - transportation sector (50% of anthropogenic)
 - industrial sector (50%; vaporization of organics)
 - biogenic (large; isoprene and monoterpenes)

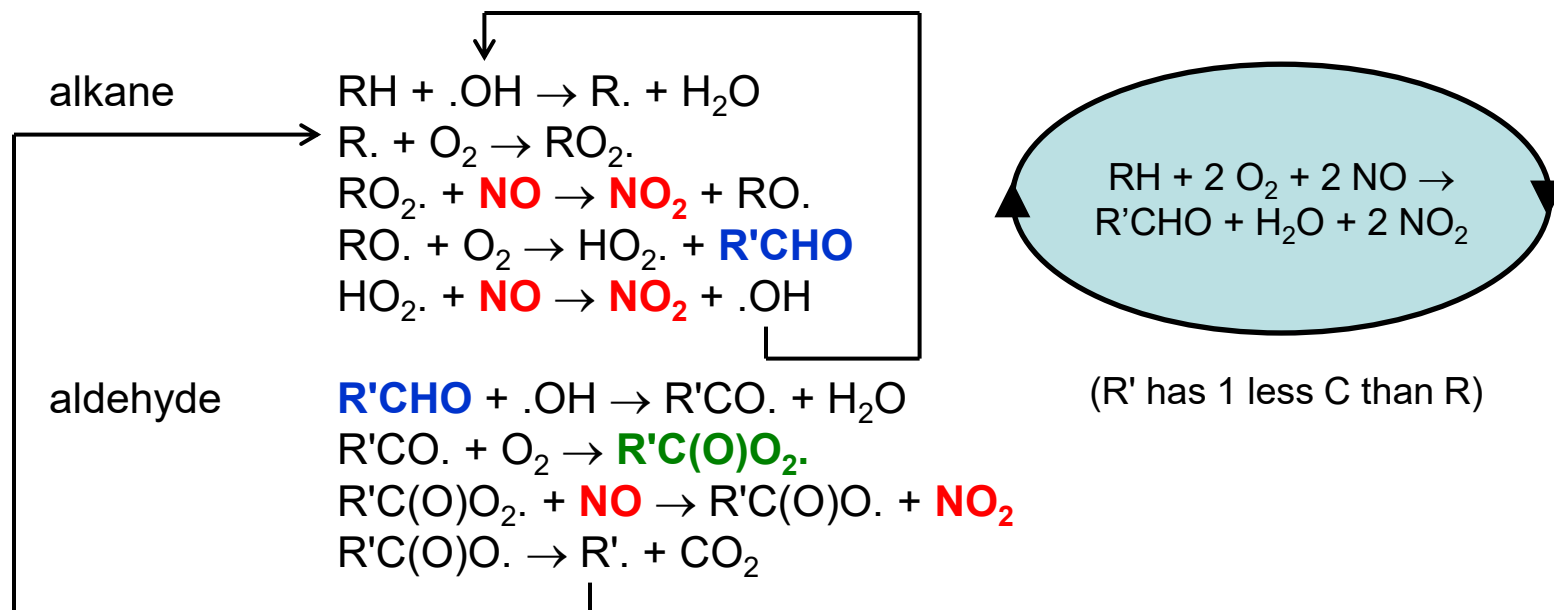


isoprene



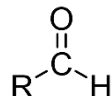
limonene

general reaction mechanism

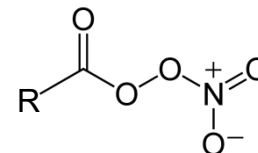


NO converted to **NO₂**

aldehydes (**R'CHO**) are formed

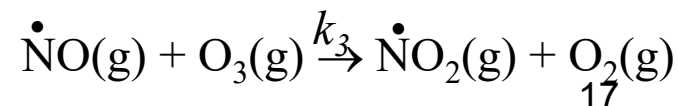
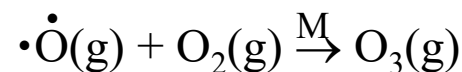
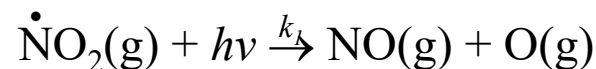
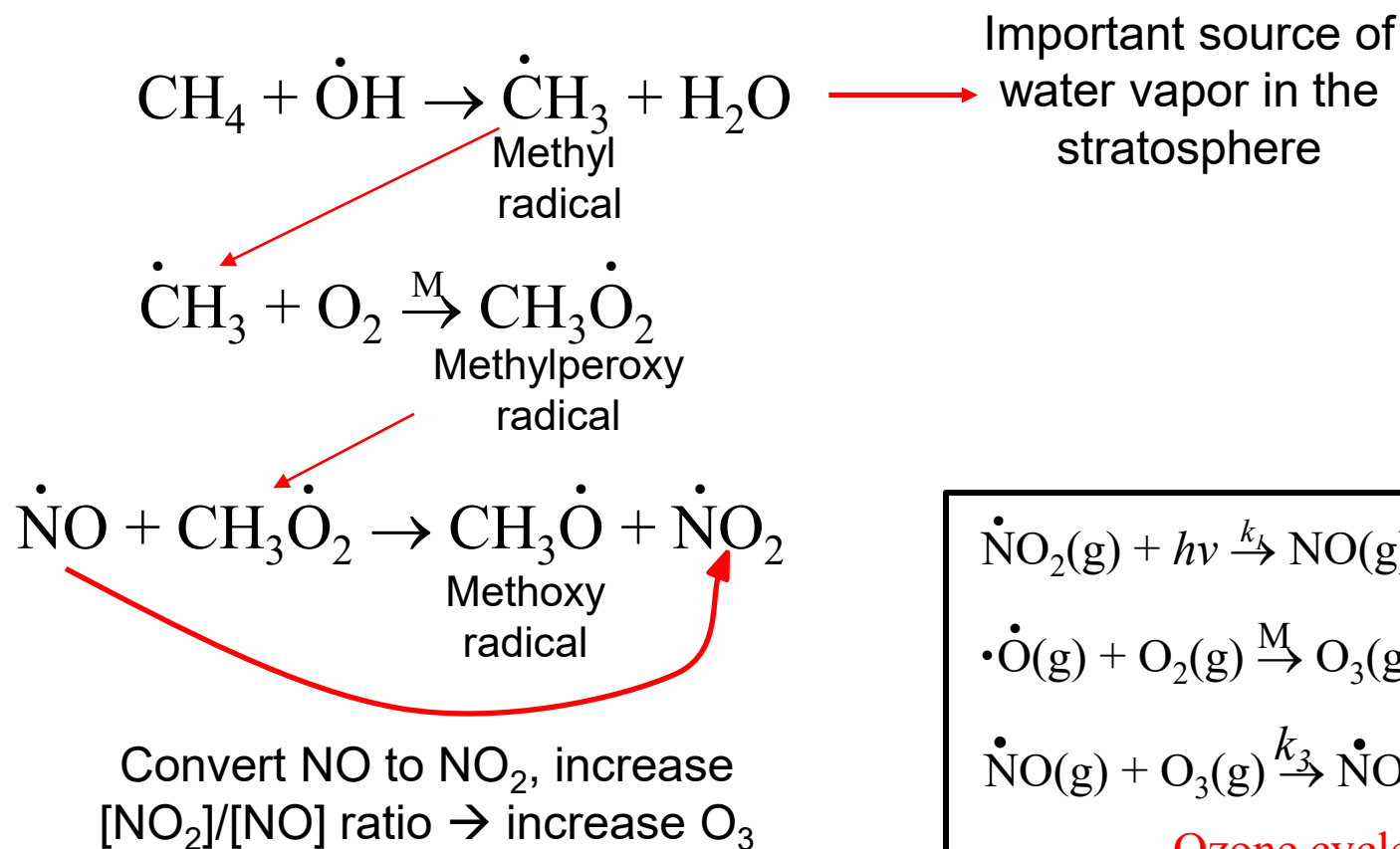


PANs (peroxyacetyl nitrates) can be formed:



Example: effect of CH₄ on O₃

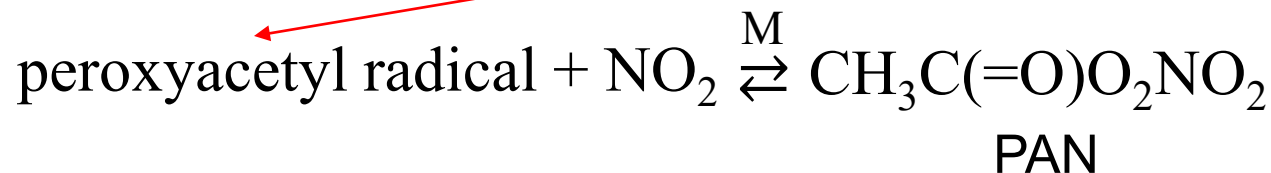
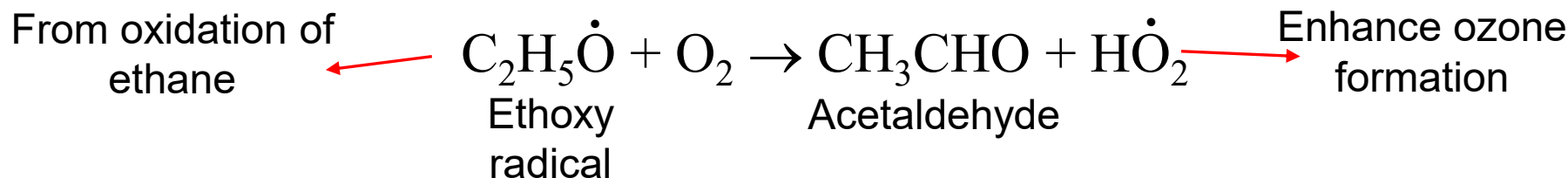
- Methane is the most abundant organic gas in the atmosphere (8-12 years of e-folding lifetime).



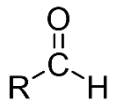
Ozone cycle

Example: PAN

- Acetaldehyde is an important product of ethane oxidation, precursor to peroxyacetyl nitrate (PAN)



relative reactivity of organics

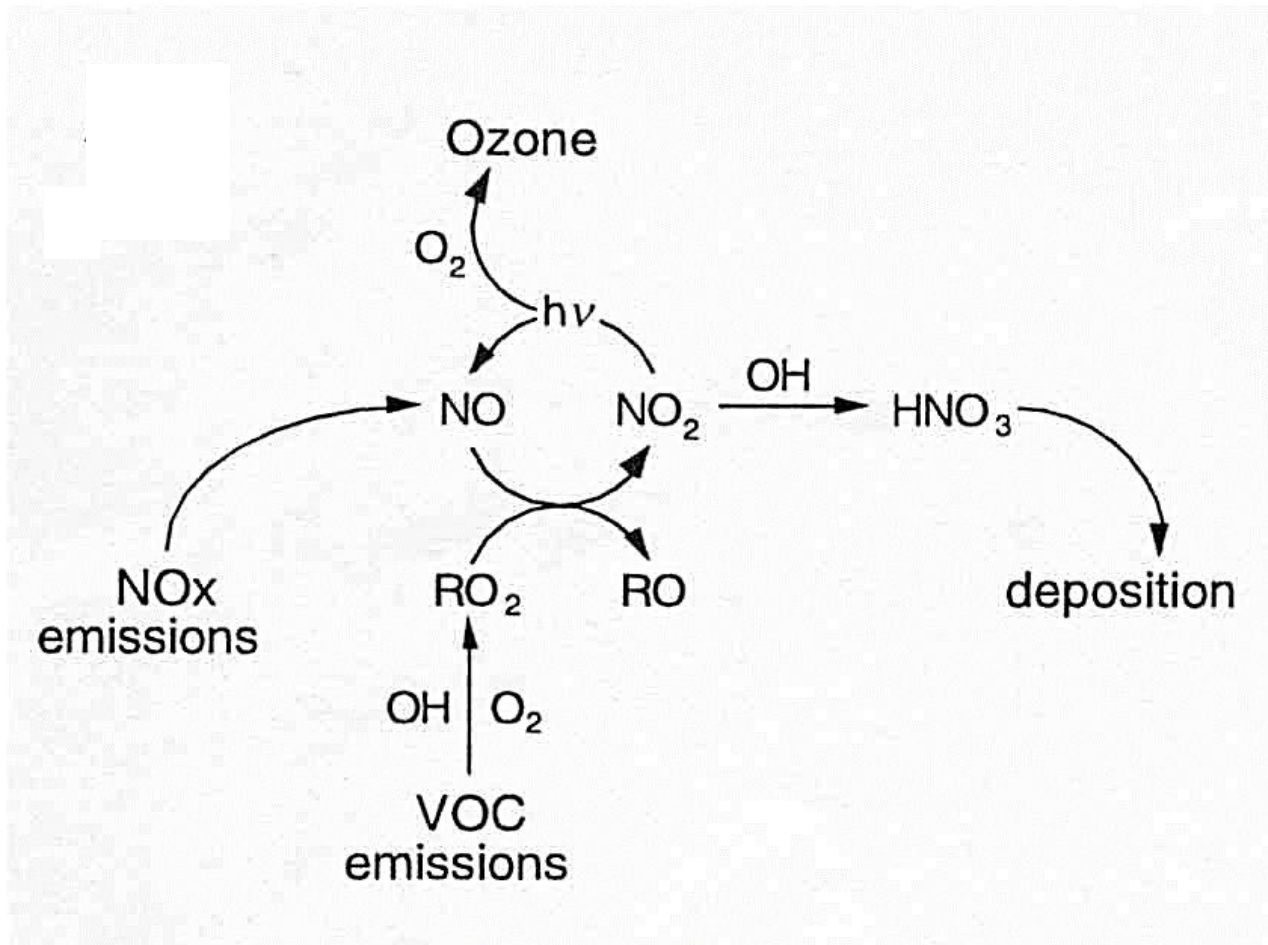
	k (25°C) cm ³ molecule ⁻¹ s ⁻¹	1/(k [.OH]) ^a
CO + .OH → CO ₂ + H.	1.5x10 ⁻¹³	36 days
CH ₄ + .OH → CH ₃ . + H ₂ O	6.3x10 ⁻¹⁵	3 years
/ C ₂ H ₆ + .OH → C ₂ H ₅ . + H ₂ O	2.6x10 ⁻¹³	30 days
< C ₃ H ₈ + .OH → C ₃ H ₇ . + H ₂ O	1.2x10 ⁻¹²	7 days
other alkanes, C _n H _{2n+2} + .OH → ...		6 hrs - 7 days
// C ₂ H ₄ + .OH → .CH ₂ -CH ₂ OH	8.5x10 ⁻¹²	22 hours
</ C ₃ H ₆ + .OH → CH ₃ -.CH-CH ₂ OH	2.6x10 ⁻¹¹	7 hours
other alkenes, C _n H _{2n} + .OH → ...		2 - 8 hours
 RCHO + .OH → RC.O + H ₂ O		2 - 12 hours

^aTime takes for reactant concentration drop to 37%, assuming
[.OH] = 0.06 ppt = 1.5x10⁶ molecules/cm³

Role of ROG in O₃ formation

- ROG enhance O₃ formation by converting NO to NO₂, but it takes time for their effects to kick in
- While wind carries the mixture of NO_x and ROG away, reactions occur that increase O₃ concentration.
- Highest ozone concentration not found near NO_x emission sources (urban regions), but rather downwind
- ROG reactions produce aldehydes, many are human carcinogens

Urban ozone cycle



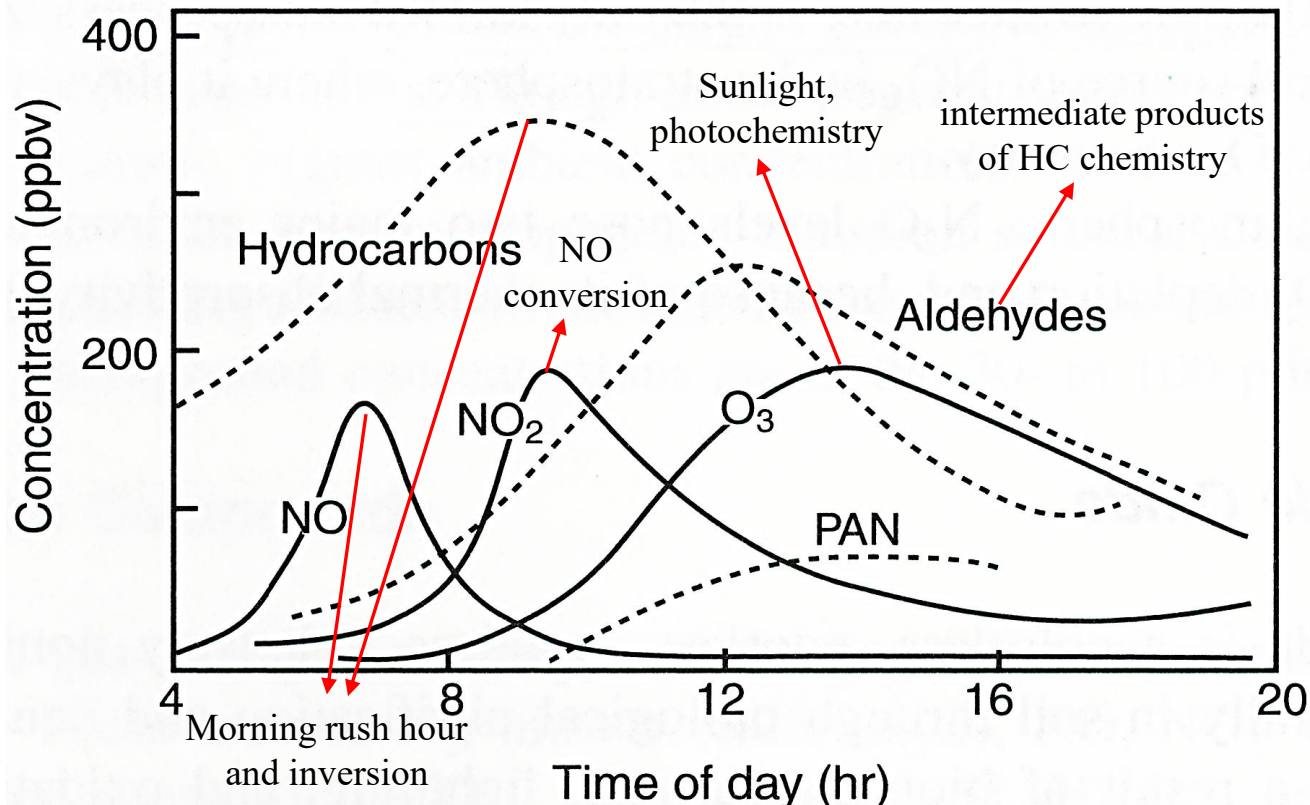
Ryerson, T. B., et al. (2001) *Science*, 292:719-722.

Urban ozone cycle

- Ozone formation potential of different VOCs differs depending on
 - Abundance of reactant
 - Rate of initial attack on precursor (by OH)
 - # of $\text{NO} \rightarrow \text{NO}_2$ conversions per VOC molecule
 - Number of cycles
 - Potential of ROG as an NO_2 sink
 - e.g. formation of PAN

Diurnal variations in smog

Three important factors: Emissions, Mixing height, Chemistry

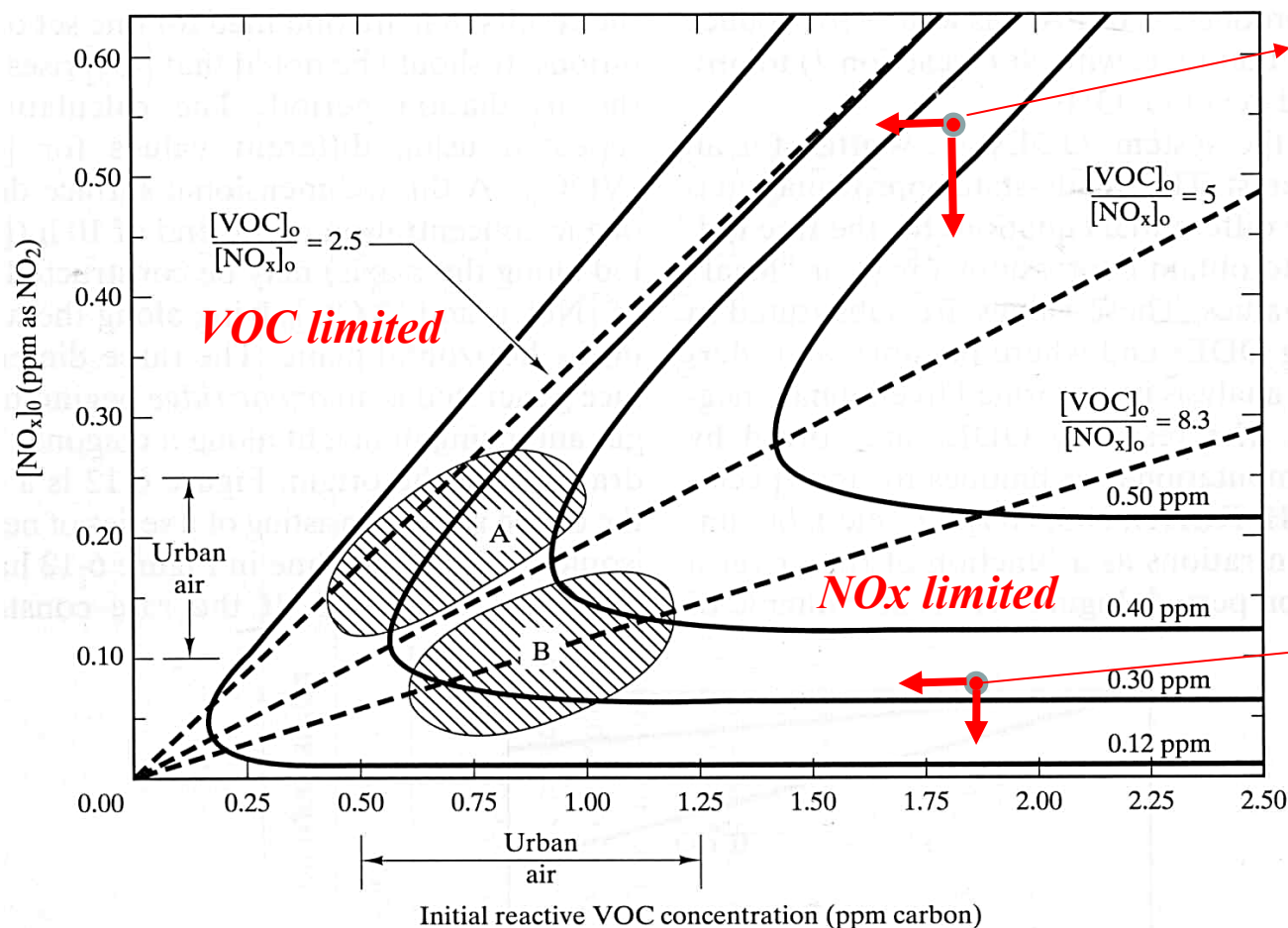


Typical pollutant concentration variation during one day (Godish, 2003)

Control of ozone

- As a secondary pollutants, ozone concentration typically reaches max at 50-100 km downwind
- Reduction in NO_x/VOC emissions does not necessarily reduce ozone concentration
- Depending on where you are at the ozone isopleth
 - Ozone concentration $\sim$ VOC & NO_x concentration

Ozone isopleth



Reduction in NO_x
increase O_3 .
 Reduction in VOC
decrease O_3 .

Reduction in NO_x
decrease O_3 .
 Reduction in VOC
no effect on O_3 .

peroxyacetyl nitrates

- Important component of photochemical smog
 - Damage plants, eye irritant, powerful tear gas
 - “Explosive” reaction
- Potential serious health effects
- Stable when cold, but quickly decomposed when hot
 - Last for long time when rises to upper atmosphere
 - Can be transported long distance (can be global)
 - Quickly decomposed if brought down near surface